

Finals Essay Task 1.

Assessment on your Understanding as to what really matters to become Data Analysts.

And How to integrate SQL into your Data Analysis

Instruction:

Read the attached Article and video link before answering the

questions. Video link: <https://youtu.be/hGCKdMsfU7g>

Article Link:

<https://drive.google.com/file/d/1FhanZgD5YFiBJP0088UbEWaN7DEFbgiE/view?usp=sharing>

Following the video that you have watch and article that I ask you to read please answer the ff: question using your own words in 2–3 sentences based on the provided source material.

1. What are the four primary factors that will differentiate a data analyst candidate in 2026?

- Based on the video, four attributes that will make a data analyst stand out in 2026 are mastery of a given domain, effective communication and storytelling abilities, visible portfolio with tangible evidence of execution, and problem-solving AI-tools that can be viewed as an asset, as opposed to a shortcut. Knowing the tools is no longer sufficient, companies are seeking an individual who will be able to connect the data with actual business decisions. When a candidate has all four of these, he or she stands out much higher than one who only concentrates on technical ability.

2. How does domain knowledge provide a competitive advantage over purely technical candidates?

- Domain knowledge helps analysts understand the real-world meaning behind the data and ask better questions. It allows them to give relevant insights that align with business goals, not just technical outputs. This makes their analysis more valuable to decision makers.

3. What is the "lean tech stack" recommended for a foundational data analyst role?

- The lean tech stack is kept really simple, just Excel, SQL, and either Power BI or Tableau to get started. Python is actually not required for most foundational roles, which is honestly a relief for beginners who feel overwhelmed trying to learn everything at once. The whole point is to get really good at a small set of tools rather than spreading yourself too thin trying to master all of them.

4. Distinguish between "learning projects" and "showing projects."

- Learning projects are basically your messy behind-the-scenes practice work where you

are still figuring things out, and those are not meant to be shared with anyone. Showing projects, on the other hand, are the polished pieces in your portfolio that actually prove you can solve real business problems and not just finish a tutorial. The difference really matters because hiring managers want to see actual business impact, not just proof that you completed a course.

5. According to the roadmap, why is Python often deprioritized for mid-career data analyst positions?

- The roadmap is constructed on efficiency - finding out what you need that will make you get hired rather than what superficially seems impressive. Your resume is already equipped with Excel, SQL and a BI tool, in most of the data analyst job descriptions, so Python on top of that will not benefit you much in resolving your job inevitable, especially in terms of time wastage. Especially to career changers, it is much more sensible to refine your knowledge of your domain and communication skills first as these are less easily taught and more valuable to employers in the first place anyway.

Questions on SQL-Base on the Article Posted

6. Why is establishing the "Business Context" essential for an analyst? What is a Business Context?

- Business context means understanding why you are looking at the data and what problem the company is trying to solve. Without it, you might write a perfect query but answer the wrong question entirely. It is essential because it makes sure your analysis is actually useful and points in the right direction.

7. According to the text, how does SQL function as a "flashlight"?

- SQL is like a flashlight since it allows you to see what you can not see in a database. It never makes the information, it just aids you in locating and disclosing it. It is the person holding the flashlight who is the analyst and it is equally important to know where to point it as it is to know how to use the flashlight.

8. Why is the validation step considered a professional necessity rather than an optional task?

- Validation ensures that the results of a query are accurate and reliable before being used for decisions. Mistakes in data analysis can lead to wrong conclusions and business risks. Professionals must always verify their work to maintain credibility.

9. What defines "good SQL" according to the author's standards? - Good SQL is not similar to code that works it is code that is clean, easy to read and too it answers the question of the business. It must be designed such that it can be looked at by another person in a way that it can be known what it is doing. It is not to impress anybody with the intricate code but write something small and useful.

10. Describe the fundamental mindset shift required to move from being a student to being an analyst.

- The transition is made based on no longer acquiring tools but addressing real-life problems. Analysts focus on delivering insights and making decisions, not just completing tasks. It requires thinking critically, asking the right questions, and taking ownership of results.